SUDEEP MULLAGURI

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SKILLS

- **Programming Languages:** Python, C, C++, Embedded C, Assembly Language, Verilog.
- **Embedded Systems:**STM32, ESP32, Arduino, 8051 Microcontroller, FreeRTOS, UART, SPI, I2C, GPIO, PWM, ADC, Timers, Interrupts, DMA, Firmware Development
- **PCB Design & Electronics:** EasyEDA, Proteus, Cadence, PSpice, Circuit Design, PCB Layout Design, Schematic Design
- **CAD & Simulation:** AutoCAD, Creo, Fusion 360, MATLAB, Wokwi, TinkerCAD
- **Tools:**Git, GitHub, Keil uVision, STM32CubeIDE, NC-SIM
- **Soft Skills:** Leadership, Problem Solving, Team Collaboration, Adaptability

TRAINING

Next-Gen Computer Networking: May'25 –July'25

- Gained hands-on experience with advanced networking concepts including VLANs, inter-VLAN routing, and network security.
- Worked with Cisco Packet Tracer to design, configure, and simulate modern enterprise networks.
- Learned and applied protocols such as DHCP, STP, and OSPF to optimize network performance and scalability.
- Built a solid foundation in next-generation networking concepts with practical exposure to real-world applications.

PROJECTS

Gesture-Controlled Bot using Computer Vision March'25

- Built a real-time gesture-controlled robot using a laptop webcam and computer vision techniques.
- Implemented hand gesture recognition using **MediaPipe** and **OpenCV** in Python.
- Converted recognized hand gestures into movement commands such as forward, left, right, and stop.
- Transmitted control commands wirelessly from the laptop to **ESP32 over Wi-Fi.**
- Programmed the ESP32 to receive commands and control motors for smooth robot movement

NanoDrive 0.1 – Custom ESP Pico D4 Development Board & PCB Design October'24

- Engineered and fabricated a custom development board based on the **ESP Pico D4**.
- Optimized PCB layout for stability, modularity, and maker-focused use cases.
- Enabled robotics and IoT prototyping with reduced form factor and cost.

ORIVA – AI-Based Smart Irrigation System December'24

- Created an AI-powered irrigation system for predictive irrigation scheduling
- Integrated ESP32, BLE, and soil moisture sensing for real-time field monitoring.
- Built a React Native application for calibration, monitoring, and irrigation recommendations.
- Leveraged weather forecasts and soil water balance modeling to predict crop water requirements.
- Reduced water wastage through intelligent irrigation recommendations.

CERTIFICATES

- Next-Gen Computer Networking (LPU) July'25
- Introduction to Python Course (Scalar) April'24

ACHIEVEMENT

- **Winner – GNA Hackathon 4.0** for developing an innovative solution in Agriculture, Food, and Rural Development.
- **Runner-Up – Infineon Battery Management System (BMS) Hackathon** for designing an advanced battery monitoring and management solution.

EDUCATION

Lovely Professional University Punjab,India

Bachelor of Technology –(ECE) **CGPA: 7.2** Aug' 23-Present

Narayana Jr. college AP, India

Intermediate; **Percentage:97%** Apr'21 – Mar'23

Narayana EM School AP,India

Matriculation; **Percentage:98%** Apr'20 –Mar'21